Harmonic Numbers

On the p - divisibility of H_n

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Introduction

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$n \mid \frac{a}{b} \iff n \mid a$ and write $\frac{a}{b} \equiv 0 \pmod{n}$.
$\frac{a}{b} \equiv \frac{c}{d} \pmod{n} \iff \frac{a}{b} - \frac{c}{d} \equiv 0 \pmod{n}$



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The p-**valuation** of an integer n is highest exponent of p which divides n.

- $\nu_3(9) = 2$
- $\nu_3(4) = 0$
- $\nu_3(\frac{4}{9}) = -2$

More Definitions and Properties

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For all primes $p \geq 5$, $p^2 \mid H_{p-1}$. So, $p - 1 \in J_p^{(1)}$.



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Theorem (Eswarathasan and Levine, 1989)

For $r \in \{0, 1, \ldots, p-1\}$, $pn + r \in J_p \iff n \in J_p$ and $H_r + \frac{1}{p} H_n \equiv 0 \pmod{p}$



Example of J_p

Recall

$J_p = \{n \in \mathbb{N} : p \mid H_n\}$
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Let's see this in action with $p = 11$:



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Let's see this in action with $p = 11$:

$J_{11}^{(1)}$	$J_{11}^{(2)}$	$J_{11}^{(3)}$	$J_{11}^{(4)}$
3	77	848	9328
7	80	852	9331
10	84	856	9335
	87	882	9338
	110	888	9376
	113	958	9378
	117	962	10583
	120	966	10587
		1291	10591
		1293	14205
			14207

Figure: First 32 elements of J_{11}



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Consider $J_p \cap [x, x+y] = \{n_1, \ldots, n_c\}$. For an integer k, let $a(k) = \#\{j : n_{j+1} - n_j = k\}$.

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$J_p(x) \leq 129 p^{2/3} x^{0.765}$ *(Sanna)*



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$J_p(x) \leq 129p^{2/3}x^{0.765}$ *(Sanna)*
$J_p(x) \leq 3x^{2/3+1/(25\log p)}$ *(Wu and Chen)*



A Fact of Our Own

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Computations suggested $\frac{p-1}{2} \in J_p \Rightarrow \lfloor \frac{p-1}{4} \rfloor \in J_p$. In looking for a proof we found this result:



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Idea

We proved this by showing $H(p-1) - \frac{1}{2} H(\frac{p-1}{2}) \equiv A(\frac{p-1}{2}) \pmod{p}$, where $A(n) = \sum_{k=1}^{n} \frac{(-1)^{k+1}}{k}$.



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Question

What is the density of $\mathcal{Q}_p$ and $\mathcal{L}$, i.e. the 'proportion' of natural numbers in $\mathcal{Q}_p, \mathcal{L}$?
To answer this, we need to know what's in $\mathcal{Q}_p$



Elements of $\mathcal{Q}_{p,m}$

Theorem (Shiu, 2016)

$n \in \mathcal{Q}_p \iff n \in [p^k r, p^k r + p^k)$, *for* $k \geq 1$ *and* $r \in J_p^{(1)}$



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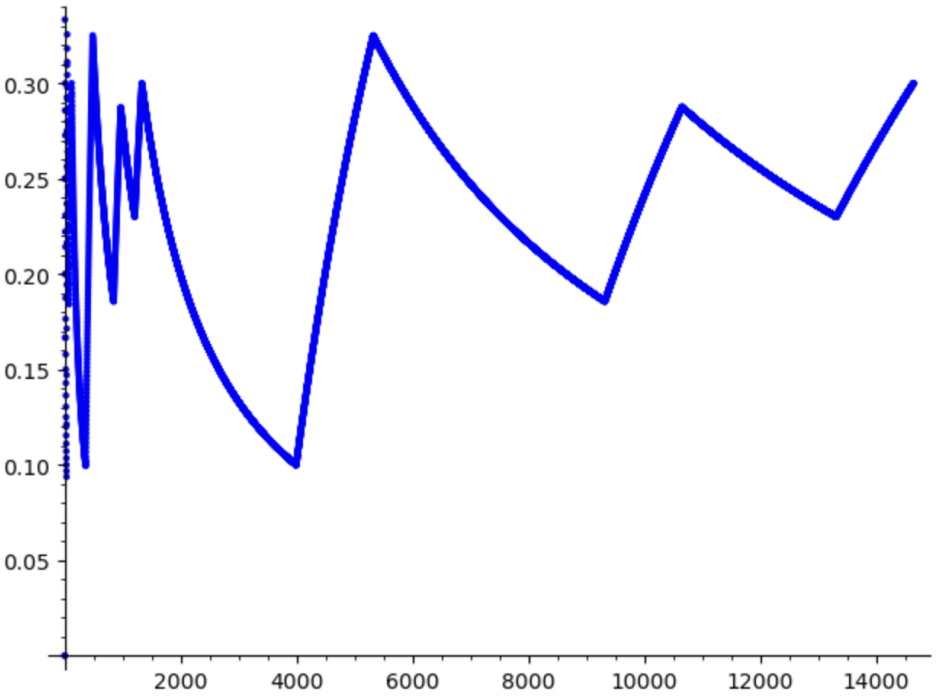
Observation

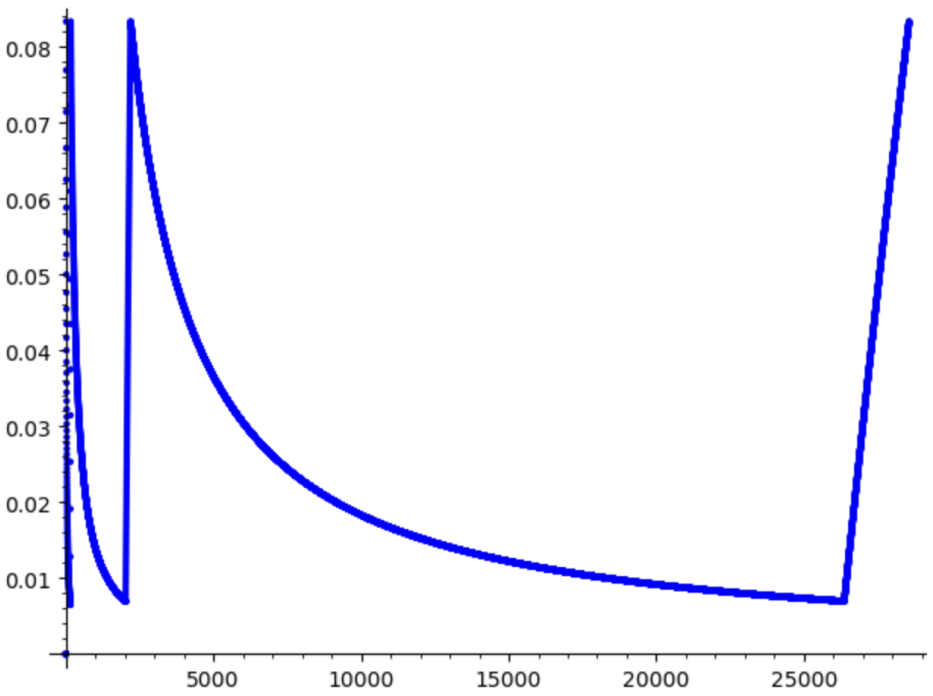
Intervals of this form are disjoint. Observe this in the following graphs $\rightarrow$



The Density of $\mathcal{Q}_p$

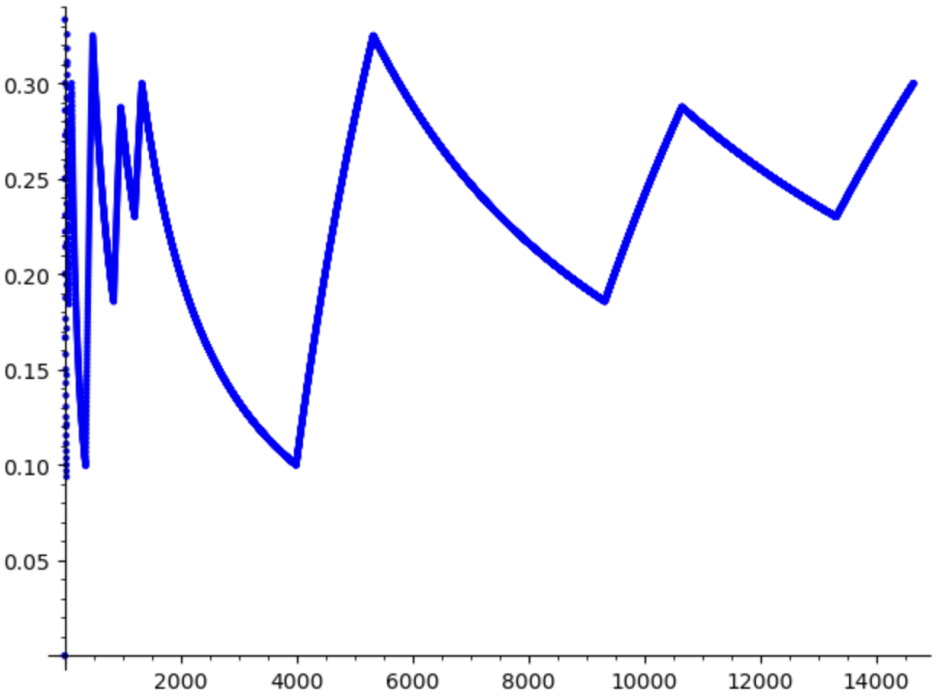
Here are graphs of the functions $\frac{\mathcal{Q}_{11} \cap [1,x]}{x}$ and $\frac{\mathcal{Q}_{13} \cap [1,x]}{x}$.

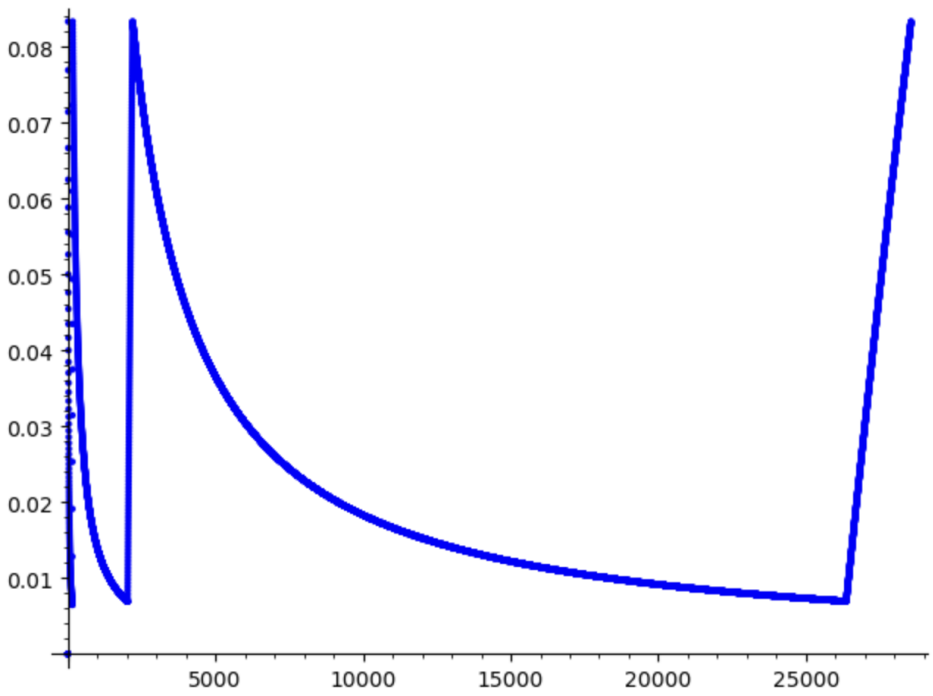




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- $\lim_{x \to \infty} \frac{Q_{11}(x)}{x}$ doesn't exist. The harmonic density does.
- We can look at lim sup and lim inf

The Density of $\mathcal{Q}_p$ (cont.)

- Recall $n \in \mathcal{Q}_p \iff n \in [p^k r, p^k r + p^k - 1],\ r \in J_p^{(1)}$
- Look at the beginnings and ends for minimal and maximal values
- $J_p^{(1)} = \{r_1, \ldots, r_s\}$



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Supremum: Take $x_{k,j} = p^k r_j + p^k - 1$. Then

$$\limsup \frac{\mathcal{Q}_p(x)}{x} = \max_{1 \le j \le s} \left\{ \frac{\frac{|J_p^{(1)}|}{p-1} + j}{r_j + 1} \right\}$$



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Infimum: Take $x_{k,j} = p^k r_j$. Then

$$\liminf \frac{\mathcal{Q}_p(x)}{x} = \min_{1 \le j \le s} \left\{ \frac{\frac{|J_p^{(1)}|}{p-1} + j - 1}{r_j} \right\}$$



lim sup Examples

$$J_{29}^{(1)} = \{13, 15, 28\}$$



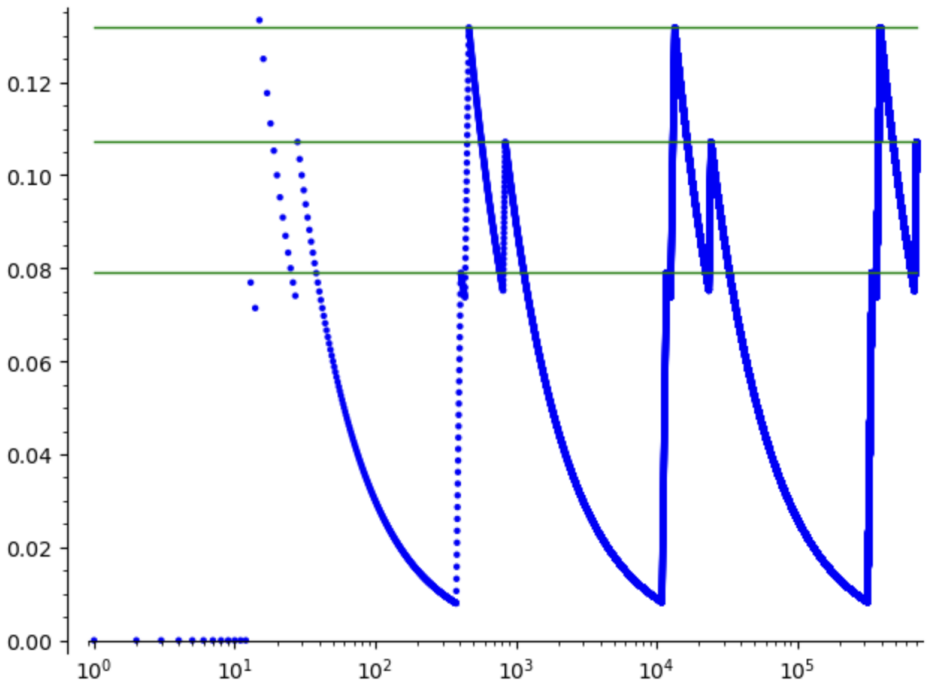
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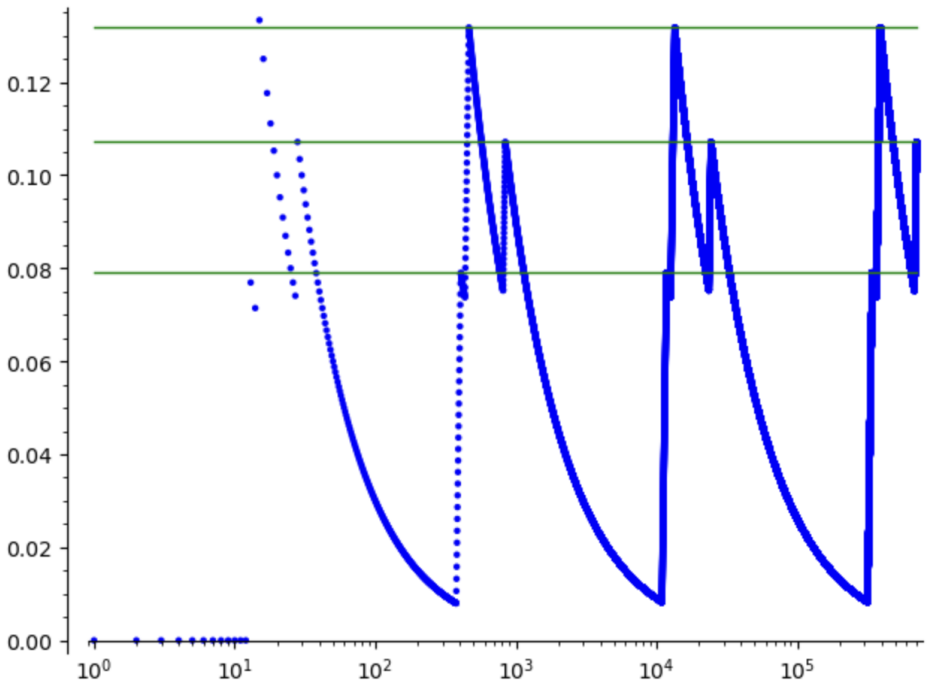
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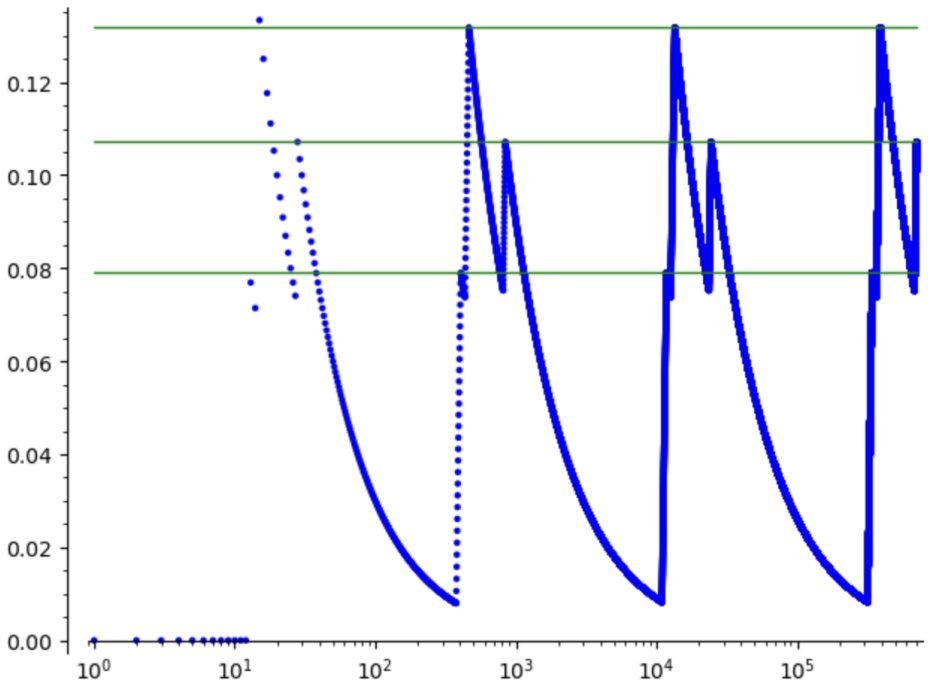


$$J_{31}^{(1)} = \{30\}$$



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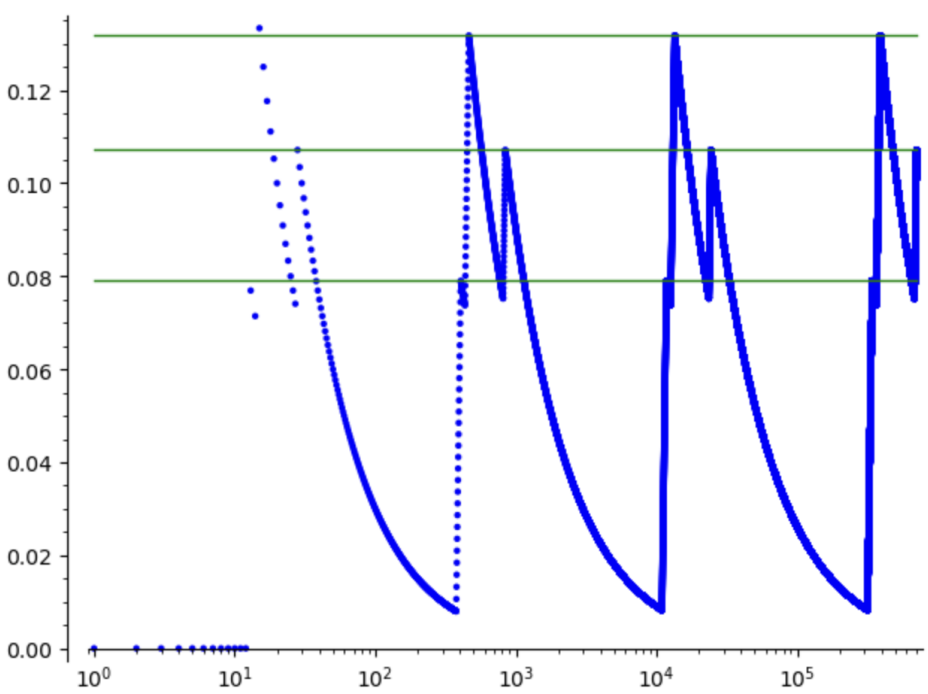
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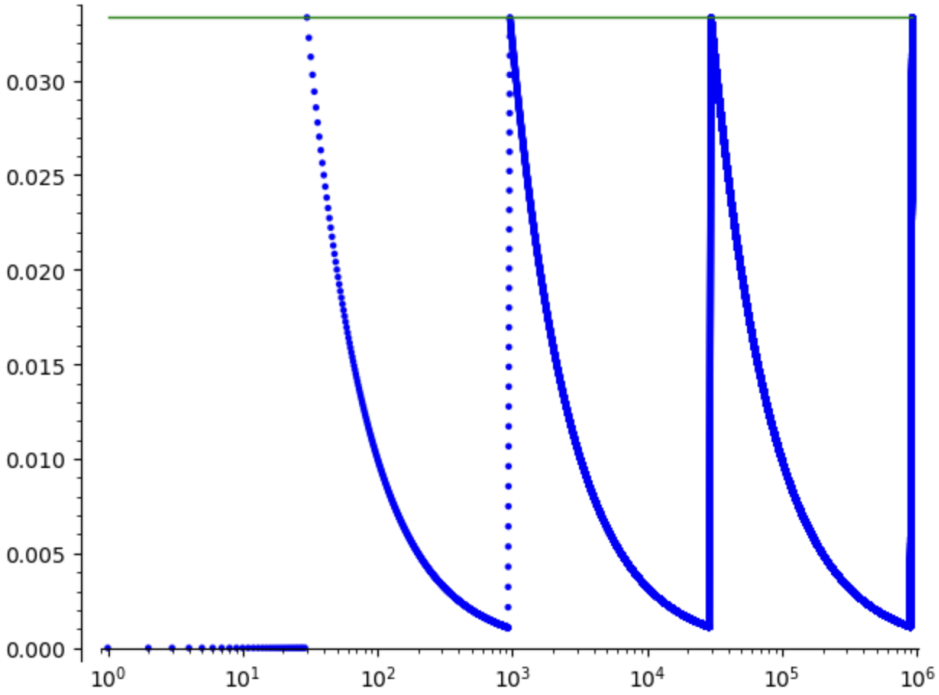
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bye bye :)

